

Theory of Change

Purpose

The purpose of this file is to analyze the data collected from the survey and to present the findings in a clear and concise manner. The analysis will include descriptive statistics, visualizations, and any relevant insights that can be drawn from the data.

Data Ingestion

```
# Load necessary libraries
```

```
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.5.2

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.1.4      v readr      2.1.5
```

```
v forcats   1.0.0      v stringr    1.5.1
```

```
v ggplot2    4.0.0      v tibble     3.2.1
```

```
v lubridate  1.9.4      v tidyr      1.3.1
```

```
v purrr      1.0.4
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(readxl)
```

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
# Read the survey data
```

```
# survey_data <- readxl::read_excel("data/Consumer Behavior towards Organic Food Comparative
```

```
survey_data <- readxl::read_excel("F:/playground/The-Theory-of-Cognitive-Affective-Mediation
```

```
# View the structure of the data
```

```
survey_data |>  
  tibble()
```

```
# A tibble: 262 x 78
```

	Timestamp <dtm>	Select your language / Wybierz j-1 <chr>	Do you agree to part-2 <chr>
1	2025-11-01 12:36:24	<NA>	I have read and under~
2	2025-11-02 16:22:30	<NA>	I have read and under~
3	2025-11-03 09:26:02	<NA>	I have read and under~
4	2025-12-15 12:00:53	Polish	<NA>
5	2025-12-17 11:53:15	English	I have read and under~
6	2025-12-17 16:05:34	English	I have read and under~
7	2025-12-17 18:28:54	English	I have read and under~
8	2025-12-17 19:38:16	English	I have read and under~
9	2025-12-17 20:32:55	English	I have read and under~
10	2025-12-17 22:27:51	English	I have read and under~

```
# i 252 more rows
```

```
# i abbreviated names: 1: `Select your language / Wybierz język`,
```

```
# 2: `Do you agree to participate in this study after reading the information above?`
```

```
# i 75 more variables:
```

```
# `Czy po zapoznaniu się z powyższymi informacjami wyraża Pan/Pani zgodę na udział w tym ba
```

```
# `A1. Have you purchased or consumed any organic food products in the last 12 months?` <cl
```

```
# `A1. Czy w ciągu ostatnich 12 miesięcy kupował (a) Pan/Pani lub spożywał(a) produkty eko
```

Data Cleaning

```
dim(survey_data)
```

```
[1] 262  78
```

```
# survey_data |>

# filter(!is.na(select_your_language_wybierz_jezyk))|>

# count(select_your_language_wybierz_jezyk)

# Define the mapping as a named list

coalesce_map <- list(
  consent = c("Do you agree to participate in this study after reading the information above?",
               "Czy po zapoznaniu się z powyższymi informacjami wyraża Pan/Pani zgodę na udział w badaniu?"),
  purchased_organic_last_12m = c("A1. Have you purchased or consumed any organic food products in the last 12 months?",
                                  "A1. Czy w ciągu ostatnich 12 miesięcy kupiłeś/kupiłaś lub zjadłeś/zjadłaś produkty organiczne?"),
  age = c('A2. Age:', 'A2. Wiek:'),
  gender = c('A3. Gender', 'A3. Płeć:'),
  education = c('A4. Highest Educational Level Completed:', 'A4. Najwyższy ukończony poziom wykształcenia:'),
  household_econ = c('A5. How would you describe the economic situation of your household?', 'A5. Jak byś opisał/a sytuację ekonomiczną twojego gospodarstwa domowego?'),
  employment = c('A6. Current Employment Status:', 'A6. Obecny status zatrudnienia:'),
  country = c('A7. Country of Residence:', 'A7. Kraj zamieszkania:'),
  awareness_duration = c('B1. How long have you been aware of organic food products?', 'B1. Jak długo jesteś świadomy/a produktów organicznych?'),
  pct_income_food = c('B2. Approximately what percentage of your total monthly household income do you spend on organic food?', 'B2. Około jaki procent twojego całkowitego miesięcznego dochodu gospodarstwa domowego przeznaczasz na produkty organiczne?'),
  pct_foodbudget_organic = c('B3. Of your total food budget, what percentage is typically spent on organic food?', 'B3. Z twojego całkowitego budżetu na żywność, jaki procent jest zwykle przeznaczany na produkty organiczne?'),
  purchase_frequency = c('B4. How often do you typically purchase organic food?', 'B4. Jak często zwykle kupujesz produkty organiczne?'),
  area_population = c('B5. What is the population size of your area of residence?', 'B5. Jaka jest liczba mieszkańców w twoim miejscu zamieszkania?'),
  buy_where = c('B6. Where do you most often buy organic food? (Select all that apply)', 'B6. Gdzie najczęściej kupujesz produkty organiczne? (Wybierz wszystkie, które dotyczą Ciebie)'),
  buy_for_whom = c('B7. For whom do you primarily buy organic food? (Select all that apply)', 'B7. Dla kogo głównie kupujesz produkty organiczne? (Wybierz wszystkie, które dotyczą Ciebie)'),
  identify_method = c('B8. What is your primary method for identifying organic products?', 'B8. Jaką metodą najczęściej używasz do rozpoznawania produktów organicznych?'),
  c1_organic_belief = c('Part 1: Your Beliefs About Organic Food\r\nPlease indicate your level of agreement with the following statement: I believe organic food is better for my health.', 'Część 1: Twoje przekonania o żywności organicznej\r\nProszę wskazać poziom zgody z następującym stwierdzeniem: Wierzę, że żywność organiczna jest lepsza dla mojego zdrowia.'),
  c2_harmful_chemicals_belief = c('Part 1: Your Beliefs About Organic Food\r\nPlease indicate your level of agreement with the following statement: I believe organic food is free from harmful chemicals.', 'Część 1: Twoje przekonania o żywności organicznej\r\nProszę wskazać poziom zgody z następującym stwierdzeniem: Wierzę, że żywność organiczna jest wolna od szkodliwych chemikaliów.'),
  c3_environment_belief = c('Part 1: Your Beliefs About Organic Food\r\nPlease indicate your level of agreement with the following statement: I believe organic food is better for the environment.', 'Część 1: Twoje przekonania o żywności organicznej\r\nProszę wskazać poziom zgody z następującym stwierdzeniem: Wierzę, że żywność organiczna jest lepsza dla środowiska.'),
  c4_ethics_belief = c('Part 1: Your Beliefs About Organic Food\r\nPlease indicate your level of agreement with the following statement: I believe organic food is fairer to the people who produce it.', 'Część 1: Twoje przekonania o żywności organicznej\r\nProszę wskazać poziom zgody z następującym stwierdzeniem: Wierzę, że żywność organiczna jest fair dla ludzi, którzy ją produkują.'),
  d1_organic_emotions = c("Please indicate your level of agreement with the following statement: I feel more confident about my health when I eat organic food.", "Proszę wskazać poziom zgody z następującym stwierdzeniem: Czuję się bardziej pewny/a w kwestii swojego zdrowia, gdy jem/am żywność organiczną."),
  d2_assurance_emotion = c("Please indicate your level of agreement with the following statement: I feel more assured about my food choices when I buy organic food.", "Proszę wskazać poziom zgody z następującym stwierdzeniem: Czuję się bardziej pewny/a w swoich wyborach dotyczących żywności, gdy kupuję żywność organiczną."),
  d3_satisfaction_emotion = c("Please indicate your level of agreement with the following statement: I feel more satisfied with my food choices when I buy organic food.", "Proszę wskazać poziom zgody z następującym stwierdzeniem: Czuję się bardziej zadowolony/a z swoich wyborów dotyczących żywności, gdy kupuję żywność organiczną."),
  d4_health_worry_emotion = c("Please indicate your level of agreement with the following statement: I feel more worried about my health when I don't eat organic food.", "Proszę wskazać poziom zgody z następującym stwierdzeniem: Czuję się bardziej zaniepokojony/a w kwestii swojego zdrowia, gdy nie jem/am żywności organicznej."),
  d5_trust_emotion = c("Please indicate your level of agreement with the following statement: I feel more trusting of organic food producers when I buy organic food.", "Proszę wskazać poziom zgody z następującym stwierdzeniem: Czuję się bardziej ufny/a wobec producentów żywności organicznej, gdy kupuję żywność organiczną.")
)
```

```

cert_ZW = c("C6. Zimbabwe Organic Certification Logo\r\nHow familiar are you with this logo")
cert_familiar = c("C7. How important is seeing an official organic certification logo when buying organic food")
price_perception = c("E3. \"Organic food is too expensive for my budget.\" ", "E3. Żywność ekologiczna jest za droga")
f1_intent = c("F1. Compared to now, do you intend to increase your consumption of organic food")
)

# survey_data |> dplyr::select(starts_with("A5")) |> names()

# Coalesce columns

for (canonical in names(coalesce_map)) {
  cols <- coalesce_map[[canonical]]
  survey_data[[canonical]] <- coalesce(!!!survey_data[cols])
}

# Optionally, remove the original columns

# survey_data <- survey_data[, !(names(survey_data) %in% unlist(coalesce_map))]

# Get only Polish and Zimbabwean respondents

survey_data_clean <- survey_data |>
  clean_names() |>
  filter(country %in% c("Poland", "Polska", "Zimbabwe")) |>
  mutate(country = case_when(country %in% c("Poland", "Polska") ~ "Poland",
                              country == "Zimbabwe" ~ "Zimbabwe",
                              TRUE ~ country))

# Clean gender variable

survey_data_clean <- survey_data_clean |>
  mutate(gender_clean = case_when(
    str_to_lower(gender) %in% c("female", "kobieta", "woman") ~ "Female",
    str_to_lower(gender) %in% c("male", "mężczyzna", "mezczyzna", "man") ~ "Male",
    TRUE ~ gender))

# select only 100 polish and 100 zimbabwean respondents
set.seed(123) # For reproducibility

survey_data_sample <- survey_data_clean |>
  group_by(country) |>

```

```
slice_sample(n = 100) |>
ungroup()
```

```
required_pkgs <- c(
  "tidyverse", # data wrangling & visualisation
  "lavaan", # SEM & mediation (Ha1, Ha2)
  "semPlot", # SEM path diagrams
  "mediation", # causal mediation analysis
  "lme4", # mixed models
  "interactions", # moderation plots (Ha3)
  "sandwich", # robust SEs
  "lmtest", # coefficient tests
  "psych", # composite reliability, alpha
  "broom", # tidy model outputs
  "modelsummary", # publication tables
  "car" # VIF, Anova
)

to_install <- required_pkgs[!required_pkgs %in% installed.packages()[,"Package"]]
if (length(to_install)) install.packages(to_install, dependencies = TRUE)
invisible(lapply(required_pkgs, library, character.only = TRUE))
```

Warning: package 'lavaan' was built under R version 4.5.2

This is lavaan 0.6-21
lavaan is FREE software! Please report any bugs.

Warning: package 'semPlot' was built under R version 4.5.2

Warning: package 'mediation' was built under R version 4.5.2

Loading required package: MASS

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

select

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

Loading required package: mvtnorm

Loading required package: sandwich

Warning: package 'sandwich' was built under R version 4.5.2

mediation: Causal Mediation Analysis

Version: 4.5.1

Warning: package 'interactions' was built under R version 4.5.2

Loading required package: zoo

Attaching package: 'zoo'

The following objects are masked from 'package:base':

as.Date, as.Date.numeric

Warning: package 'psych' was built under R version 4.5.2

Attaching package: 'psych'

The following object is masked from 'package:mediation':

mediate

The following object is masked from 'package:lavaan':

cor2cov

The following objects are masked from 'package:ggplot2':

%+%, alpha

Warning: package 'modelsummary' was built under R version 4.5.2

Attaching package: 'modelsummary'

The following object is masked from 'package:psych':

SD

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:psych':

logit

The following object is masked from 'package:dplyr':

recode

The following object is masked from 'package:purrr':

some

The Theory of Cognitive-Affective Mediation

Alternative (H): Affective emotions are positively impacted by cognitive beliefs, which raise the frequency of purchases.

Null (H): Affective sentiments do not enhance the frequency of purchases, while cognitive ideas have no effect on affective feelings.

```
# =====  
  
# Define mapping for purchase frequency  
  
purchase_freq_map <- c(  
  "daily" = 5,  
  "codziennie" = 5,  
  "a few times a week" = 4,  
  "kilka razy w tygodniu" = 4,  
  "once a week" = 3,  
  "raz w tygodniu" = 3,  
  "2-3 times a month" = 2,  
  "2-3 razy w miesiącu" = 2,  
  "once a month" = 2,  
  "raz w miesiącu" = 2,  
  "less than once a month / occasionally" = 1,  
  "rzadziej niż raz w miesiącu / sporadycznie" = 1  
)  
  
# Clean and harmonize purchase frequency  
  
survey_data_clean_df <- survey_data_sample |>  
  mutate(  
    purchase_frequency_clean = str_to_lower(str_trim(purchase_frequency)),  
    purchase_frequency_clean = str_replace_all(purchase_frequency_clean, "-", "-"),  
    purchase_frequency_rank = purchase_freq_map[purchase_frequency_clean]  
  )  
  
# survey_data_clean_df %>% count(purchase_frequency_rank) |> print(n = Inf)  
  
# =====  
  
# library(dplyr)  
  
df_ha1 <- survey_data_clean_df %>%
```

```

dplyr::select(country, purchase_frequency_rank,
               c1_organic_belief:c4_ethics_belief,
               d1_organic_emotions:d5_trust_emotion)

# df_ha1 %>% dplyr::select(purchase_frequency_rank)

# # Count values per cur_column

# for (col in names(df_ha1)) {

# cat("n---", col, "value counts ---n")

# print(table(df_ha1[[col]], useNA = "ifany"))

# }

# Define Likert mapping (English + Polish)

likert_map <- c(
  "strongly disagree" = 1,
  "disagree" = 2,
  "neutral" = 3,
  "agree" = 4,
  "strongly agree" = 5,
  "zdecydowanie nie zgadzam się" = 1,
  "zdecydowanie się nie zgadzam" = 1,
  "nie zgadzam się" = 2,
  "raczej się nie zgadzam" = 2, "neutralny" = 3,
  "nie mam opinii" = 3,
  "ani się zgadzam, ani się nie zgadzam" = 3,
  "zgadzam się" = 4,
  "raczej się zgadzam" = 4,
  "strongly agree" = 5,
  "zdecydowanie zgadzam się" = 5,
  "zdecydowanie się zgadzam" = 5)

# List columns to clean

belief_emotion_cols <- c(
  "c1_organic_belief", "c2_harmful_chemicals_belief", "c3_environment_belief", "c4_ethics_be",
  "d1_organic_emotions", "d2_assurance_emotion", "d3_satisfaction_emotion", "d4_health_worry",
  "d5_trust_emotion")

```

```

# Clean and convert to numeric

df_ha1 <- survey_data_clean_df %>%
  dplyr::select(country, purchase_frequency_rank,
                c1_organic_belief, c2_harmful_chemicals_belief,
                c3_environment_belief, c4_ethics_belief,
                d1_organic_emotions, d2_assurance_emotion,
                d3_satisfaction_emotion, d4_health_worry_emotion,
                d5_trust_emotion) %>%
  mutate(across(all_of(belief_emotion_cols),
                  ~likert_map[str_to_lower(str_trim(.))], .names = "{.col}_num")) %>%
  mutate(purchase_frequency_rank = case_when(
    is.na(purchase_frequency_rank) ~ NA_integer_,
    purchase_frequency_rank == 'Invalid Number' ~ NA_integer_,
    TRUE ~ as.numeric(purchase_frequency_rank)))

# Creating composite scores for Cognitive and Affective constructs

df_ha1_df <- df_ha1 %>%
  mutate(
    Cognitive = rowMeans(cbind(
      c1_organic_belief_num, c2_harmful_chemicals_belief_num,
      c3_environment_belief_num, c4_ethics_belief_num), na.rm = TRUE),
    Affective = rowMeans(cbind(
      d1_organic_emotions_num, d2_assurance_emotion_num,
      d3_satisfaction_emotion_num, d4_health_worry_emotion_num #, d5_trust_emotion_num
    ), na.rm = TRUE))

# Baron-Kenny mediation

library(lmtest)
library(sandwich)

# Filter for complete cases

mediation_data <- df_ha1_df %>%
  dplyr::select(Cognitive, Affective, purchase_frequency_rank) %>%
  drop_na()

# Step 1: Cognitive → Purchase Frequency
# bk1 <- lm(purchase_frequency_rank ~ Cognitive, data = df_ha1_df)

```

```
bk1 <- lm(purchase_frequency_rank ~ Cognitive, data = mediation_data)
coeftest(bk1, vcov = vcovHC(bk1, "HC3"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.6376751	0.2569788	14.1555	<2e-16 ***
Cognitive	-0.0033412	0.0761116	-0.0439	0.965

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Step 2: Cognitive → Affective
# bk2 <- lm(Affective ~ Cognitive, data = df_ha1_df)
bk2 <- lm(Affective ~ Cognitive, data = mediation_data)
coeftest(bk2, vcov = vcovHC(bk2, "HC3"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.611593	0.284094	5.6727	5.843e-08 ***
Cognitive	0.540509	0.081675	6.6178	4.451e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Step 3: Cognitive + Affective → Purchase Frequency
# bk3 <- lm(purchase_frequency_rank ~ Cognitive + Affective, data = df_ha1_df)
bk3 <- lm(purchase_frequency_rank ~ Cognitive + Affective, data = mediation_data)
coeftest(bk3, vcov = vcovHC(bk3, "HC3"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.64972823	0.30133784	12.1117	<2e-16 ***
Cognitive	0.00070123	0.10067423	0.0070	0.9945
Affective	-0.00747902	0.10888566	-0.0687	0.9453

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Fit models on mediation_data Mediation analysis
```

```
library(mediation)
set.seed(2025)
med <- mediation::mediate(
  model.m = bk2,
  model.y = bk3,
  treat = "Cognitive",
  mediator = "Affective",
  boot = TRUE,
  sims = 2000,
  boot.ci.type = "bca")
```

Running nonparametric bootstrap

```
summary(med)
```

Causal Mediation Analysis

Nonparametric Bootstrap Confidence Intervals with the BCa Method

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME	-4.0425e-03	-7.3614e-02	1.9707e-01	0.974
ADE	7.0123e-04	-2.5766e-01	1.4868e-01	0.936
Total Effect	-3.3413e-03	-1.4963e-01	1.4247e-01	0.950
Prop. Mediated	1.2099e+00	8.6906e+14	8.7950e+14	0.948

Sample Size Used: 174

Simulations: 2000

```
# structural Equation Modelling

#+ d5_trust_emotion_num

library(lavaan)

ha1_sem <- '
```

```

Cog =~ c1_organic_belief_num + c2_harmful_chemicals_belief_num + c3_environment_belief_num +
Aff =~ d1_organic_emotions_num + d2_assurance_emotion_num + d3_satisfaction_emotion_num + d4_

Aff ~ a * Cog
purchase_frequency_rank ~ b * Aff + cp * Cog
indirect := a * b
total := cp + (a * b)
,

# Prepare SEM data with all item columns

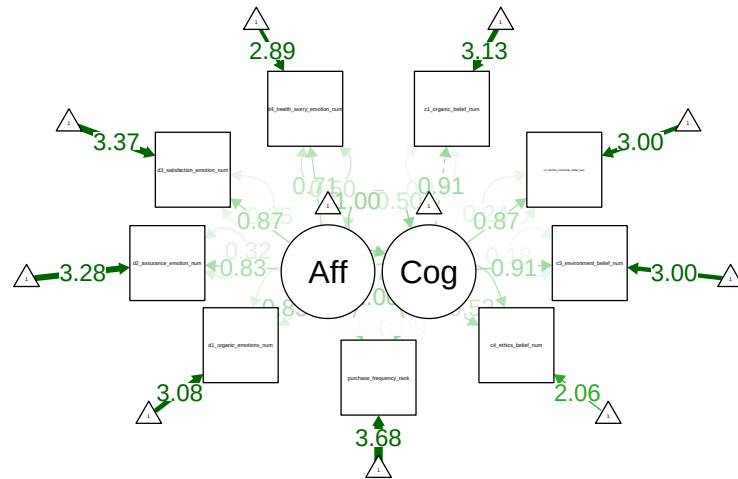
sem_data <- df_ha1_df %>%
  dplyr::select(
    c1_organic_belief_num, c2_harmful_chemicals_belief_num,
    c3_environment_belief_num, c4_ethics_belief_num,
    d1_organic_emotions_num, d2_assurance_emotion_num,
    d3_satisfaction_emotion_num, d4_health_worry_emotion_num, #d5_trust_emotion_num
    Cognitive, Affective, purchase_frequency_rank
  ) %>%
  drop_na()

# Fit SEM model

ha1_fit <- sem(ha1_sem,
  data = sem_data,
  estimator = "ML",
  se = "bootstrap",
  bootstrap = 2000,
  missing = "ml")

library(semPlot)
semPaths(ha1_fit, what = "std",
  layout = "circle",
  edge.label.cex = 1.2, nCharNodes = 0,
  sizeMan = 8,
  sizeLat = 10, title = FALSE)

```



Fitted Measures

```
fitMeasures(ha1_fit,
             c("cfi", "tli", "rmsea", "srmr", "aic", "bic"))
```

cfi	tli	rmsea	srmr	aic	bic
0.962	0.945	0.092	0.053	3943.209	4034.822

```
summary(ha1_fit, standardized = TRUE, ci = TRUE)
```

```
lavaan 0.6-21 ended normally after 28 iterations
```

Estimator	ML
Optimization method	NLMINB
Number of model parameters	29
Number of observations	174
Number of missing patterns	1

Model Test User Model:

Test statistic	61.827
----------------	--------

Degrees of freedom	25
P-value (Chi-square)	0.000

Parameter Estimates:

Standard errors	Bootstrap
Number of requested bootstrap draws	2000
Number of successful bootstrap draws	2000

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
Cog =~						
c1_rgnc_blf_nm	1.000				1.000	1.000
c2_hrmfl_chm__	0.955	0.057	16.619	0.000	0.842	1.075
c3_nvrnmnt_bl_	1.003	0.057	17.647	0.000	0.897	1.117
c4_thcs_blf_nm	0.687	0.092	7.440	0.000	0.498	0.870
Aff =~						
d1_rgnc_mtns_n	1.000				1.000	1.000
d2_ssrnc_mtn_n	0.885	0.068	13.018	0.000	0.751	1.018
d3_stsfctn_mt_	0.941	0.079	11.894	0.000	0.798	1.113
d4_hlth_wrry__	0.840	0.087	9.607	0.000	0.674	1.023
Std.lv	Std.all					
	1.059	0.915				
	1.011	0.869				
	1.062	0.905				
	0.727	0.531				
	0.964	0.846				
	0.853	0.826				
	0.907	0.869				
	0.810	0.709				

Regressions:

	Estimate	Std.Err	z-value	P(> z)	ci.lower
Aff ~					
Cog (a)	0.646	0.092	7.016	0.000	0.468
purchase_frequency_rank ~					
Aff (b)	-0.042	0.152	-0.278	0.781	-0.259
Cog (cp)	0.084	0.137	0.617	0.537	-0.262
ci.upper	Std.lv	Std.all			
	0.833	0.710	0.710		

0.339	-0.041	-0.041
0.272	0.089	0.091

Intercepts:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
.c1_rgnc_blf_nm	3.621	0.085	42.801	0.000	3.454	3.787
.c2_hrmfl_chm__	3.494	0.085	40.930	0.000	3.328	3.661
.c3_nvrnmnt_bl_	3.523	0.088	39.957	0.000	3.356	3.701
.c4_thcs_blf_nm	2.816	0.102	27.702	0.000	2.621	3.023
.d1_rgnc_mtns_n	3.506	0.087	40.482	0.000	3.333	3.678
.d2_ssrrnc_mtn_n	3.391	0.077	43.849	0.000	3.241	3.540
.d3_stsfctn_mt_	3.517	0.078	45.006	0.000	3.362	3.678
.d4_hlth_wrry__	3.305	0.084	39.399	0.000	3.138	3.471
.prchs_frqncy_r	3.626	0.073	49.559	0.000	3.471	3.764
Std.lv	Std.all					
3.621	3.129					
3.494	3.004					
3.523	3.003					
2.816	2.057					
3.506	3.080					
3.391	3.285					
3.517	3.371					
3.305	2.894					
3.626	3.683					

Variances:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
.c1_rgnc_blf_nm	0.218	0.081	2.705	0.007	0.073	0.384
.c2_hrmfl_chm__	0.331	0.079	4.172	0.000	0.183	0.493
.c3_nvrnmnt_bl_	0.249	0.061	4.071	0.000	0.135	0.376
.c4_thcs_blf_nm	1.345	0.148	9.099	0.000	1.038	1.619
.d1_rgnc_mtns_n	0.367	0.079	4.643	0.000	0.215	0.530
.d2_ssrrnc_mtn_n	0.339	0.067	5.083	0.000	0.214	0.476
.d3_stsfctn_mt_	0.267	0.056	4.777	0.000	0.155	0.377
.d4_hlth_wrry__	0.648	0.108	6.019	0.000	0.448	0.864
.prchs_frqncy_r	0.965	0.118	8.148	0.000	0.716	1.184
Cog	1.120	0.136	8.227	0.000	0.855	1.382
.Aff	0.461	0.140	3.295	0.001	0.206	0.744
Std.lv	Std.all					
0.218	0.163					
0.331	0.245					
0.249	0.181					

1.345	0.718
0.367	0.284
0.339	0.318
0.267	0.245
0.648	0.497
0.965	0.995
1.000	1.000
0.496	0.496

Defined Parameters:

	Estimate	Std.Err	z-value	P(> z)	ci.lower	ci.upper
indirect	-0.027	0.107	-0.256	0.798	-0.160	0.250
total	0.057	0.078	0.735	0.462	-0.096	0.207
Std.lv	Std.all					
-0.029	-0.029					
0.060	0.061					

How to restructure?

```
modificationIndices(ha1_fit, sort = TRUE, maximum.number = 10)
```

	lhs	op	rhs	mi
79	d4_health_worry_emotion_num	~~	purchase_frequency_rank	11.872
69	c4_ethics_belief_num	~~	purchase_frequency_rank	11.193
66	c4_ethics_belief_num	~~	d2_assurance_emotion_num	7.833
59	c3_environment_belief_num	~~	c4_ethics_belief_num	7.520
43	Aff	==	c4_ethics_belief_num	5.997
44	c1_organic_belief_num	~~	c2_harmful_chemicals_belief_num	5.452
52	c2_harmful_chemicals_belief_num	~~	c3_environment_belief_num	5.006
57	c2_harmful_chemicals_belief_num	~~	d4_health_worry_emotion_num	4.571
38	Cog	==	d3_satisfaction_emotion_num	4.412
70	d1_organic_emotions_num	~~	d2_assurance_emotion_num	4.227
	epc	sepc.lv	sepc.all	sepc.nox
79	-0.220	-0.220	-0.278	-0.278
69	-0.294	-0.294	-0.258	-0.258
66	-0.163	-0.163	-0.241	-0.241
59	0.154	0.154	0.266	0.266
43	-0.369	-0.356	-0.260	-0.260
44	0.115	0.115	0.429	0.429
52	-0.110	-0.110	-0.383	-0.383
57	0.089	0.089	0.192	0.192
38	0.166	0.176	0.169	0.169
70	0.093	0.093	0.264	0.264

The Theory of the Certification Trust Chain

Alternative (\$H_1\$): Purchase likelihood is increased by certification familiarity, which then boosts confidence after the purchase. Null (\$H_0\$): buy experience does not boost post-buy confidence, and certification familiarity does not raise purchase likelihood.

```
# purchase likelihood (F1. "I intend to increase my purchase of organic food in the next 6 months")
# certification familiarity (C5. EU Organic Logo (Used in Poland and the EU) How familiar are you with this logo?)
# confidence after purchase (D2: "I feel confident and reassured when I see an official organic logo")

df_ha2_df <- survey_data_sample |>
  dplyr::rename(
    c5_eu_logo_familiarity= c5_eu_organic_logo_used_in_poland_and_the_eu_how_familiar_are_you_with_this_logo,
    zim_logo_familiarity = cert_zw,
    organic_logo_importance = cert_familiar) |>
  dplyr::select(f1_intent, zim_logo_familiarity, organic_logo_importance, c5_eu_logo_familiarity,
               d2_assurance_emotion)

f1_intent_map <- c(
  "No, I intend to decrease it" = 1,
  "I'm not sure" = 2,
  "No, I will maintain the same level" = 3,
  "Pozostanie na tym samym poziomie" = 3,
  "Yes, slightly" = 4,
  "Raczej tak" = 4,
  "Yes, significantly" = 5,
  "Zdecydowanie tak" = 5)

# Harmonise

df_ha2_df$f1_intent_num <- f1_intent_map[str_trim(df_ha2_df$f1_intent)]

# Logo harmonisation

logo_fam_map <- c(
  "I do not recognize this logo at all" = 1,
  "Nie, nie rozpoznaję tego logo" = 1,
  "I have seen this logo but don't know much about it" = 2,
  "Widziałem/am je, ale niewiele o nim wiem" = 2,
```

```

    "I recognize this logo and know what it means" = 3,
    "Znam to logo i wiem, co oznacza" = 3)

df_ha2_df$zim_logo_familiarity_num <- logo_fam_map[str_trim(df_ha2_df$zim_logo_familiarity)]
df_ha2_df$c5_eu_logo_familiarity_num <- logo_fam_map[str_trim(df_ha2_df$c5_eu_logo_familiarity)]

# Logo importance

logo_importance_map <- c(
  "Not important - It doesn't affect my purchasing decisions" = 1,
  "Nieważne - Nie wpływa na moje decyzje zakupowe" = 1,
  "I don't pay attention to certification logos" = 1,
  "Nie zwracam uwagi na logo certyfikatów" = 1,
  "Somewhat important - It influences my choice, but I sometimes buy uncertified" = 2,
  "Dość ważne - Wpływa na mój wybór, ale czasem kupuję produkty niecertyfikowane" = 2,
  "Very important - I look for it and usually only buy certified products" = 3)

df_ha2_df$organic_logo_importance_num <- logo_importance_map[str_trim(df_ha2_df$organic_logo_importance)]

# Mediation analysis

# Purchase likelihood (f1_intent) is increased by certification familiarity (Zim logo familiarity)

# Filter for complete cases

mediation_data_2 <- df_ha2_df %>%
  dplyr::select(f1_intent_num, c5_eu_logo_familiarity_num, zim_logo_familiarity_num,
               organic_logo_importance_num) %>%
  drop_na()

# Step 1: Certification familiarity → Purchase likelihood

bk1_2 <- lm(f1_intent_num ~ c5_eu_logo_familiarity_num + zim_logo_familiarity_num, data = mediation_data_2)
coeftest(bk1_2, vcov = vcovHC(bk1_2, "HC3"))

```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.661618	0.199878	23.3223	<2e-16 ***
c5_eu_logo_familiarity_num	-0.059085	0.085803	-0.6886	0.4922
zim_logo_familiarity_num	-0.031551	0.098535	-0.3202	0.7493

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Step 2: Certification familiarity → Organic logo importance
```

```
# bk2 <- lm(Affective ~ Cognitive, data = df_ha1_df)
```

```
bk2_2 <- lm(organic_logo_importance_num ~ c5_eu_logo_familiarity_num + zim_logo_familiarity_num, data = df_ha1_df)
coeftest(bk2_2, vcov = vcovHC(bk2_2, "HC3"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.06816	0.20438	5.2263	6.411e-07 ***
c5_eu_logo_familiarity_num	0.14012	0.10649	1.3159	0.1904
zim_logo_familiarity_num	0.45713	0.11394	4.0119	9.930e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Step 3: Certification familiarity + Organic logo importance → Purchase likelihood
```

```
bk3_2 <- lm(f1_intent_num ~ c5_eu_logo_familiarity_num + zim_logo_familiarity_num + organic_logo_importance_num, data = df_ha1_df)
coeftest(bk3_2, vcov = vcovHC(bk3_2, "HC3"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.507553	0.199104	22.6392	<2e-16 ***
c5_eu_logo_familiarity_num	-0.079296	0.091134	-0.8701	0.3858
zim_logo_familiarity_num	-0.097485	0.127760	-0.7630	0.4468
organic_logo_importance_num	0.144234	0.138797	1.0392	0.3006

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Fit models on mediation_data Mediation analysis
```

```
set.seed(2025)
med_2 <- mediation::mediate(
  model.m = bk2_2,
```

```

model.y = bk3_2,
treat = "c5_eu_logo_familiarity_num",
mediator = "organic_logo_importance_num",
boot = TRUE,
sims = 2000,
boot.ci.type = "bca")

```

Running nonparametric bootstrap

```
summary(med_2)
```

Causal Mediation Analysis

Nonparametric Bootstrap Confidence Intervals with the BCa Method

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME	2.0211e-02	-9.7970e-03	1.1398e-01	0.383
ADE	-7.9296e-02	-2.5143e-01	9.6990e-02	0.355
Total Effect	-5.9085e-02	-2.2927e-01	9.5609e-02	0.470
Prop. Mediated	-3.4206e-01	2.3179e+13	2.3458e+13	0.625

Sample Size Used: 138

Simulations: 2000

The Moderating Hypothesis of Price Barriers

Alternative (\$H_1\$): The positive correlation between affective sentiments and purchase intentions is weakened by price perception, particularly for consumers with lower incomes.

Null (\$H_0\$): Regardless of economic level, the link between affective emotions and purchase intentions is not weakened by price perception.

```

# =====
# Ha3: PRICE BARRIERS MODERATION
# Affective Emotions × Price Perception → Purchase Intention (F1)
# Moderated by Income Group (SES proxy)

```

```

# =====

# e3_zywnosc_ekologiczna_jest_dla_mnie_zbyt_droga
# e3_organic_food_is_too_expensive_for_my_budget
# Define Likert mapping (English + Polish)

likert_map <- c(
  "strongly disagree" = 1,
  "disagree" = 2,
  "neutral" = 3,
  "agree" = 4,
  "strongly agree" = 5,
  "zdecydowanie nie zgadzam się" = 1,
  "zdecydowanie się nie zgadzam" = 1,
  "nie zgadzam się" = 2,
  "raczej się nie zgadzam" = 2,
  "neutralny" = 3,
  "nie mam opinii" = 3,
  "ani się zgadzam, ani się nie zgadzam" = 3,
  "zgadzam się" = 4,
  "raczej się zgadzam" = 4,
  "strongly agree" = 5,
  "zdecydowanie zgadzam się" = 5,
  "zdecydowanie się zgadzam" = 5)

affective_emotions_list <- c(
  "d1_organic_emotions", "d2_assurance_emotion",
  "d3_satisfaction_emotion", "d4_health_worry_emotion" #, "d5_trust_emotion"
)

# Harmonize and numerically code 'employment' column

# Mapping both Polish and English values to numeric codes and English labels

employment_map <- c(
  'Employed full-time' = 1,
  'Praca na pełny etat' = 1,
  'Employed part-time' = 2,
  'Praca w niepełnym wymiarze czasu' = 2,
  'Self-employed / Freelancer' = 3,
  'Bezrobotna/ny' = 4,
  'Unemployed' = 4,

```

```

'Retired' = 5,
'Student' = 6,
'Homemaker' = 7,
'Osoła prowadząca gospodarstwo domowe' = 7
)

employment_label_map <- c(
  'Employed full-time' = 'Employed full-time',
  'Praca na pełny etat' = 'Employed full-time',
  'Employed part-time' = 'Employed part-time',
  'Praca w niepełnym wymiarze czasu' = 'Employed part-time',
  'Self-employed / Freelancer' = 'Self-employed / Freelancer',
  'Bezrobotna/ny' = 'Unemployed',
  'Unemployed' = 'Unemployed',
  'Retired' = 'Retired',
  'Student' = 'Student',
  'Homemaker' = 'Homemaker',
  'Osoła prowadząca gospodarstwo domowe' = 'Homemaker')

survey_data_sample <- survey_data_sample |>
  mutate(
    employment_num = employment_map[employment],
    employment_label = employment_label_map[employment])

# Clean and convert to numeric

df_ha3 <- survey_data_sample %>%
  dplyr::select(f1_intent, price_perception, employment_num,
    all_of(affective_emotions_list)) %>%
  mutate(across(all_of(affective_emotions_list),
    ~ likert_map[str_to_lower(str_trim(.))],
    .names = "{.col}_num"))

# Creating composite scores for Cognitive and Affective constructs

df_ha3_df <- df_ha3 %>%
  mutate(Affective = rowMeans(cbind(
    d1_organic_emotions_num, d2_assurance_emotion_num,
    d3_satisfaction_emotion_num, d4_health_worry_emotion_num #, d5_trust_emotion_num
  ), na.rm = TRUE))

# Harmonize and numerically code 'price_perception' column

```

```

# Mapping: Strongly disagree = 1, Disagree = 2, Neutral = 3, Agree = 4, Strongly agree = 5
# Polish equivalents: 'Zdecydowanie się nie zgadzam' = 1, 'Nie zgadzam się' = 2, 'Ani się zgadzam, ani nie' = 3, 'Zgadza się' = 4, 'Zdecydowanie się zgadzam' = 5
# For multiple answers, take the mean of mapped values

price_perception_map <- c(
  'Strongly disagree' = 1,
  'Disagree' = 2,
  'Neutral' = 3,
  'Agree' = 4,
  'Strongly agree' = 5,
  'Zdecydowanie się nie zgadzam' = 1,
  'Nie zgadzam się' = 2,
  'Ani się zgadzam, ani nie' = 3,
  'Zgadza się' = 4,
  'Zdecydowanie się zgadzam' = 5)

df_ha3_df <- df_ha3_df |>
  mutate(
    price_perception_num = apply(strsplit(price_perception, ','),
                                  function(x) {x <- trimws(x)
                                  vals <- price_perception_map[x]
                                  as.numeric(vals[1]) # Use only the first mapped value
                                  }))

# Check the result

# df_ha3_df |> count(household_econ, household_econ_num)

f1_intent_map <- c(
  "No, I intend to decrease it" = 1,
  "I'm not sure" = 2,
  "No, I will maintain the same level" = 3,
  "Pozostanie na tym samym poziomie" = 3,
  "Yes, slightly" = 4,
  "Raczej tak" = 4,
  "Yes, significantly" = 5,
  "Zdecydowanie tak" = 5)

# Harmonise

```

```

df_ha3_df$f1_intent_num <- f1_intent_map[str_trim(df_ha3_df$f1_intent)]

# Check the result

# df_ha3_df |> count(price_perception, price_perception_num)

# survey_data_sample |> count(employment)

# Check the result

# survey_data_sample |> count(employment, employment_num, employment_label)

# Dropping rows with NA in key variables

df_ha3_df <- df_ha3_df |> drop_na(f1_intent, Affective, price_perception_num, employment_num)

# df_ha3_df <- df_ha3_df |>
#   mutate(
#     Affective_c = Affective - mean(Affective, na.rm = TRUE),
#     E3_price_c = price_perception_num - mean(price_perception_num, na.rm = TRUE))

#   Ha3-A: Base 2-way moderation

ha3_base <- lm(f1_intent_num ~ Affective * price_perception_num, data = df_ha3_df)
cat("n--- Ha3 Base: Affective x Price -> F1_intent ---n")

```

```

n--- Ha3 Base: Affective x Price -> F1_intent ---n

```

```

print(coeftest(ha3_base, vcov = vcovHC(ha3_base, "HC3")))

```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.692031	0.612776	6.0251	8.535e-09 ***
Affective	0.230852	0.166709	1.3848	0.1677
price_perception_num	0.188599	0.200636	0.9400	0.3484
Affective:price_perception_num	-0.061403	0.056014	-1.0962	0.2744

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
cat(sprintf("R² = %.4f", summary(ha3_base)$r.squared))
```

R² = 0.0168

```
# Ha3-B: 3-way with income group
```

```
ha3_3way <- lm(f1_intent_num ~ Affective * price_perception_num * employment_num, data = df_l)
cat("n--- Ha3 3-Way: Affective * Price * Employment → F1_intent ---n")
```

n--- Ha3 3-Way: Affective * Price * Employment → F1_intent ---n

```
print(coeftest(ha3_3way, vcov = vcovHC(ha3_3way, "HC3")))
```

t test of coefficients:

	Estimate	Std. Error	t value
(Intercept)	3.587951	1.423087	2.5212
Affective	0.192424	0.365300	0.5268
price_perception_num	0.100384	0.493533	0.2034
employment_num	0.011535	0.345988	0.0333
Affective:price_perception_num	-0.011961	0.129156	-0.0926
Affective:employment_num	0.019324	0.095903	0.2015
price_perception_num:employment_num	0.040365	0.119951	0.3365
Affective:price_perception_num:employment_num	-0.020357	0.034111	-0.5968

Pr(>|t|)

(Intercept)	0.01253 *
Affective	0.59899
price_perception_num	0.83904
employment_num	0.97344
Affective:price_perception_num	0.92631
Affective:employment_num	0.84053
price_perception_num:employment_num	0.73686
Affective:price_perception_num:employment_num	0.55137

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
cat(sprintf("R² = %.4f ΔR² = %.4fn",
            summary(ha3_3way)$r.squared,
            summary(ha3_3way)$r.squared - summary(ha3_base)$r.squared))
```

$R^2 = 0.0317$ $\Delta R^2 = 0.0149$

```
# Ha3-C: Model comparison`  
  
cat("n--- Ha3: ANOVA model comparison ---n")
```

n--- Ha3: ANOVA model comparison ---n

```
print(anova(ha3_base, ha3_3way))
```

Analysis of Variance Table

```
Model 1: f1_intent_num ~ Affective * price_perception_num  
Model 2: f1_intent_num ~ Affective * price_perception_num * employment_num  
   Res.Df    RSS Df Sum of Sq    F Pr(>F)  
1     191 103.89  
2     187 102.32  4     1.5696 0.7172 0.5812
```

Hypothesis of Country-Specific Demographic Drivers

Alternative (\$H_1\$): In both nations, socioeconomic position has a favorable impact on cognitive beliefs and purchase frequency, but it does so through distinct mechanisms—directly in Zimbabwe and indirectly through education and urban residency in Poland.

Null (\$H_0\$): Neither country's cognitive views nor purchase frequency are significantly impacted by socioeconomic position.

Socioeconomic position (SEP) is defined by: 1. A5 — Household Economic Situation 2. A4 — Highest Educational Level Completed 3. A6 — Current Employment Status

```
# =====  
# Ha4: COUNTRY-SPECIFIC DEMOGRAPHIC DRIVERS  
# Zimbabwe: SES → Cognitive / Purchase directly  
# Poland: SES → (Educ + Urban) → Cognitive → Purchase (indirect pathway)  
# =====  
  
# Harmonize and numerically code 'household_econ' column  
  
# # Mapping: Poor = 1, Fair/Average/Przeciętna = 2, Good/Dobra = 3, Very good = 4
```

```

household_econ_map <- c(
  'Poor' = 1,
  'Dobra' = 3,
  'Good' = 3,
  'Fair/Average' = 2,
  'Przeciętna' = 2,
  'Very good' = 4)

# zim_data <- survey_data_sample %>%
#   filter(country == "Zimbabwe") %>%
#   mutate(household_econ_num = household_econ_map[household_econ]) %>% names()
#   select(household_econ, education,
#          Cognitive, purchase_frequency_rank)

```